

AVION YF-52 Coyote

Preliminary Design Review



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Objective

Objective: Design a carrier-capable supersonic strike aircraft that **breaks cost trends**.

Performance Requirements:

- Mach **1.6-2.0** high altitude dash capability
- Carrier compatible configuration
- TOGW under **90,000 lbs**
- **700-1000 nm.** combat radius

Cut costs lower than:

- R&D cost of **\$2 trillion** for F-35 program
- Unit flyaway cost of **\$102 million** for F-35C



Overview: YF-52 Coyote

YF-52 Concept:

- R&D cost: **\$27 billion**
- Unit cost: **\$55 million**
- Gross weight: 36700 lb
- Total thrust: 33500 lb
- Engines: 1
- Wing area: 400 sq ft

F-35 Lightning II:

- R&D cost: **\$2 trillion**
- Unit cost: **\$102 million**
- Gross weight: 50000 lb
- Total thrust: 43000 lb
- Engines: 1
- Wing area: 460 sq ft

F/A-18E/F Super Hornet:

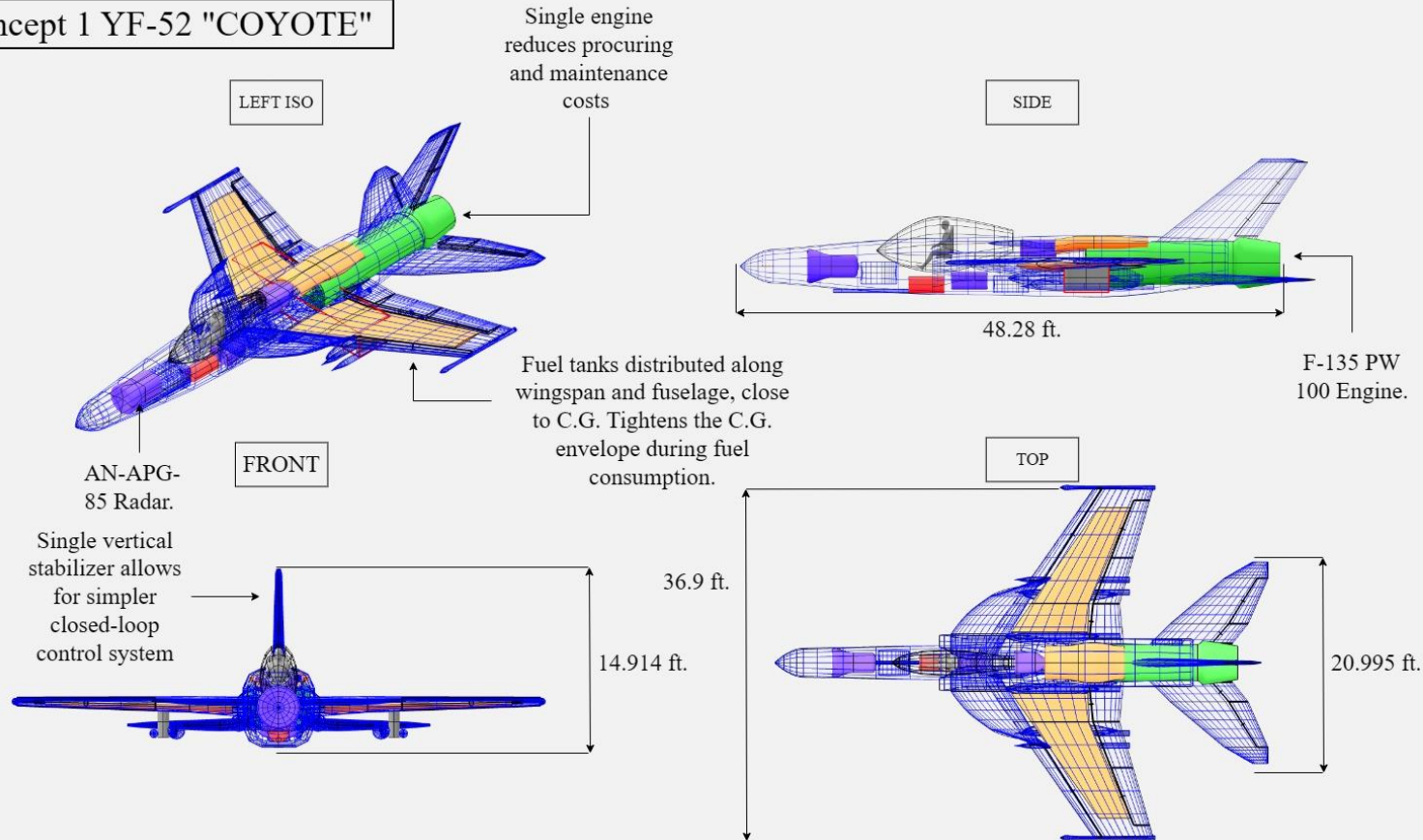
- R&D cost: **\$10 billion**
- Unit cost **\$72 million**
- Gross weight 47000 lb
- Total thrust: 44000 lb
- Engines: 2
- Wing area: 500 sq ft



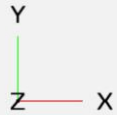
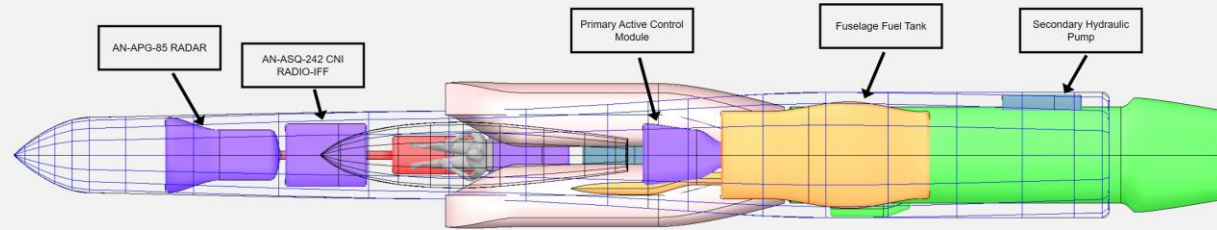
98% decrease in development cost.
46% decrease in unit cost.

Dimensioned three-view drawing of the COYOTE

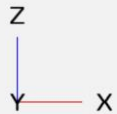
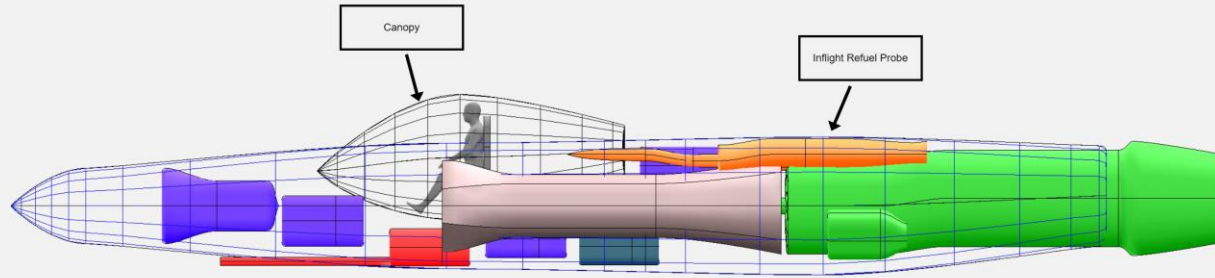
Concept 1 YF-52 "COYOTE"



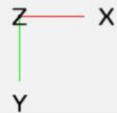
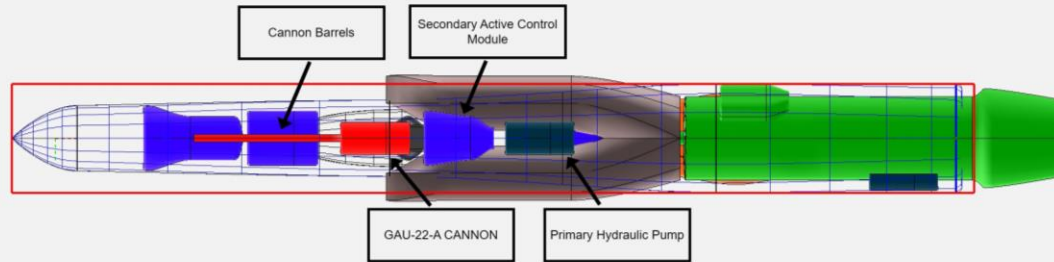
Fuselage



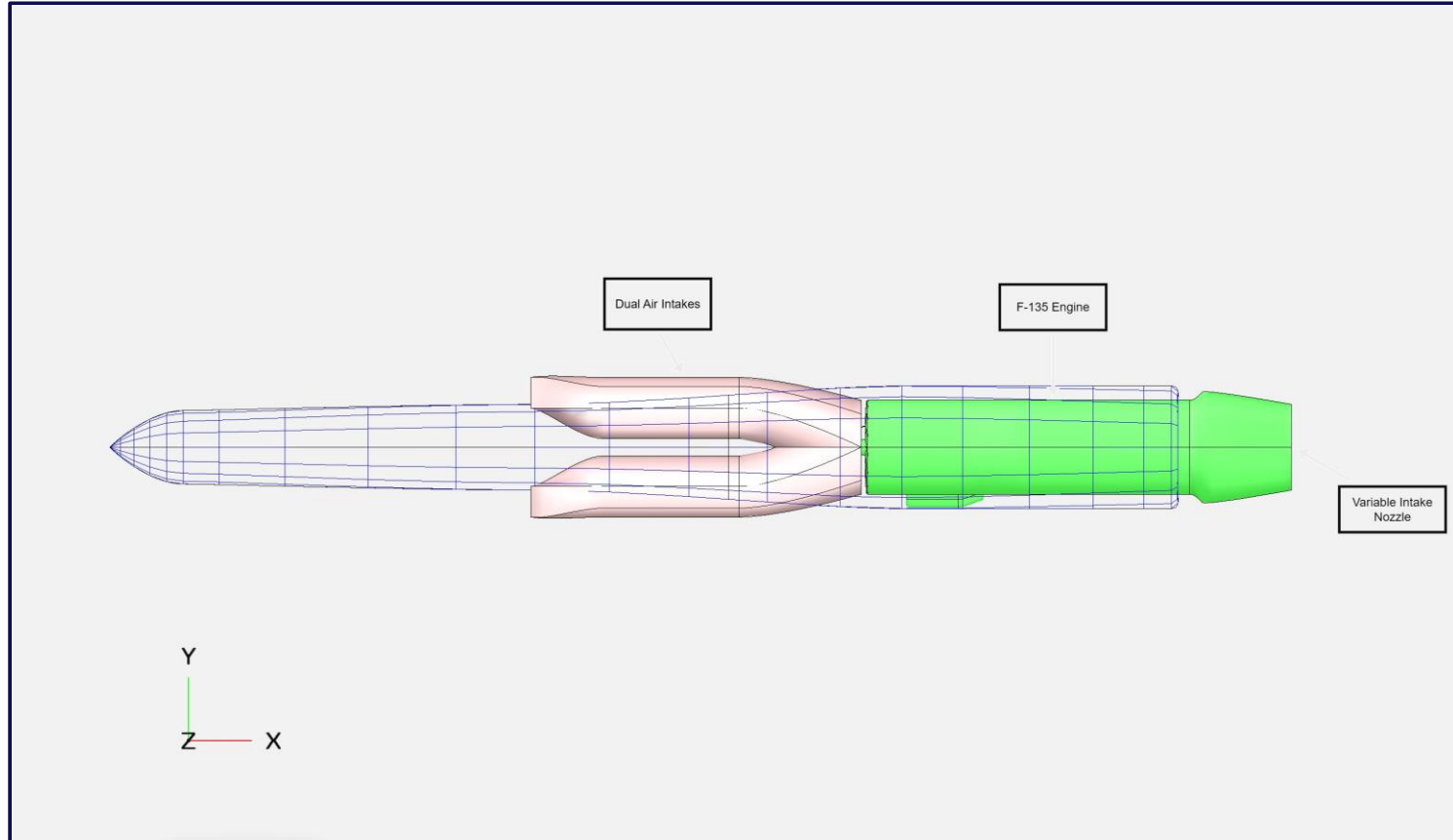
Fuselage



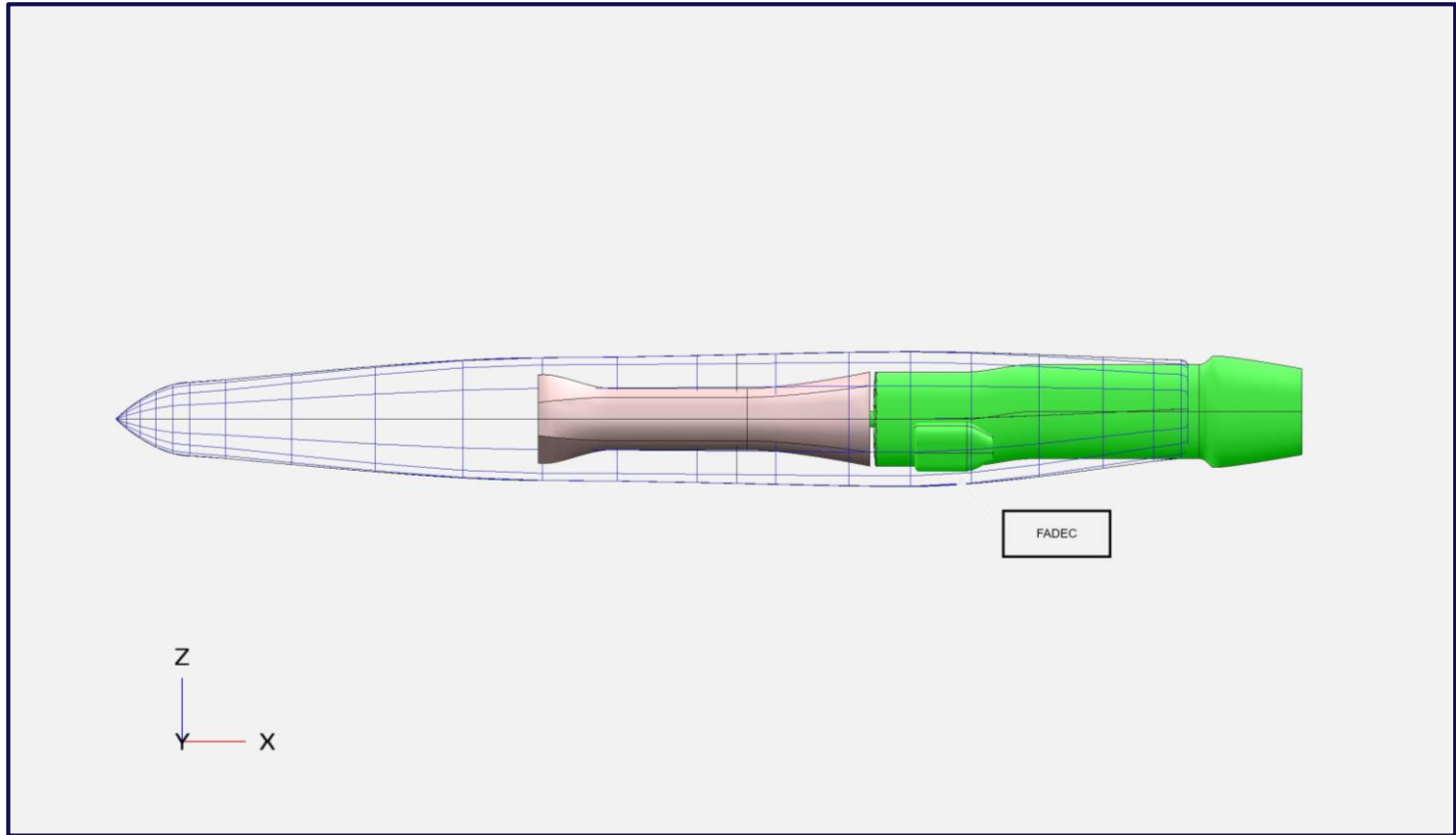
Fuselage



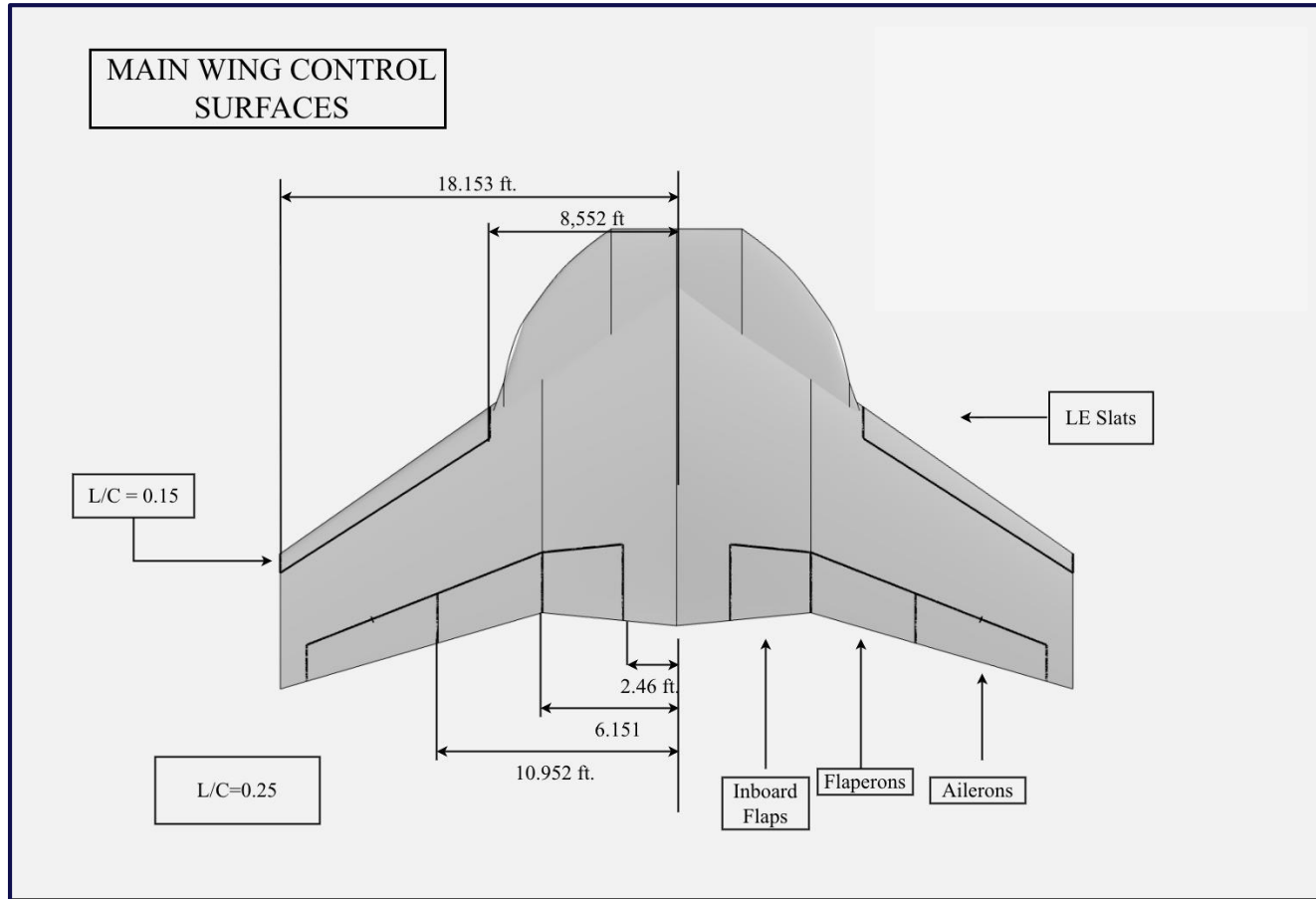
Powertrain



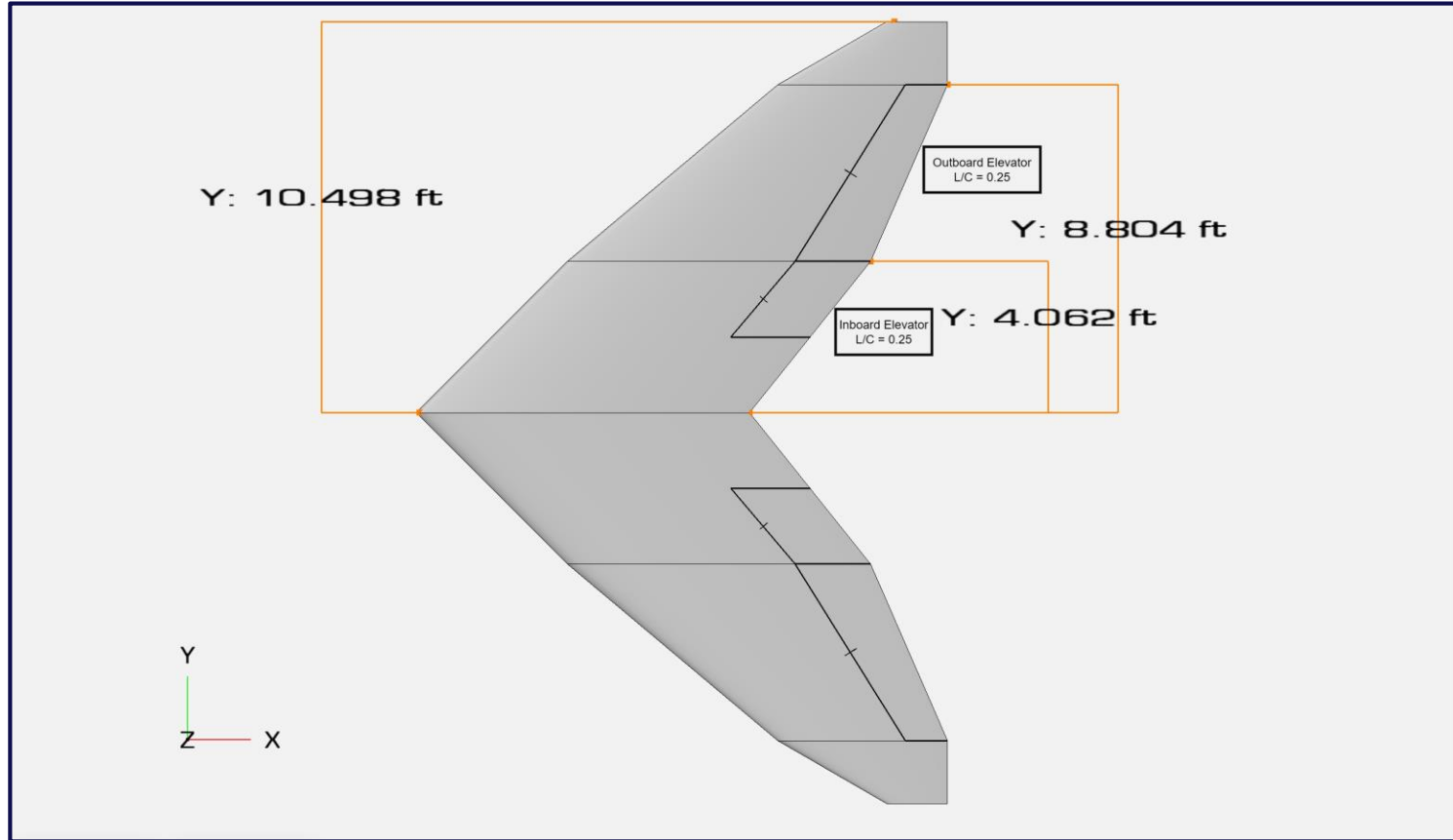
Powertrain



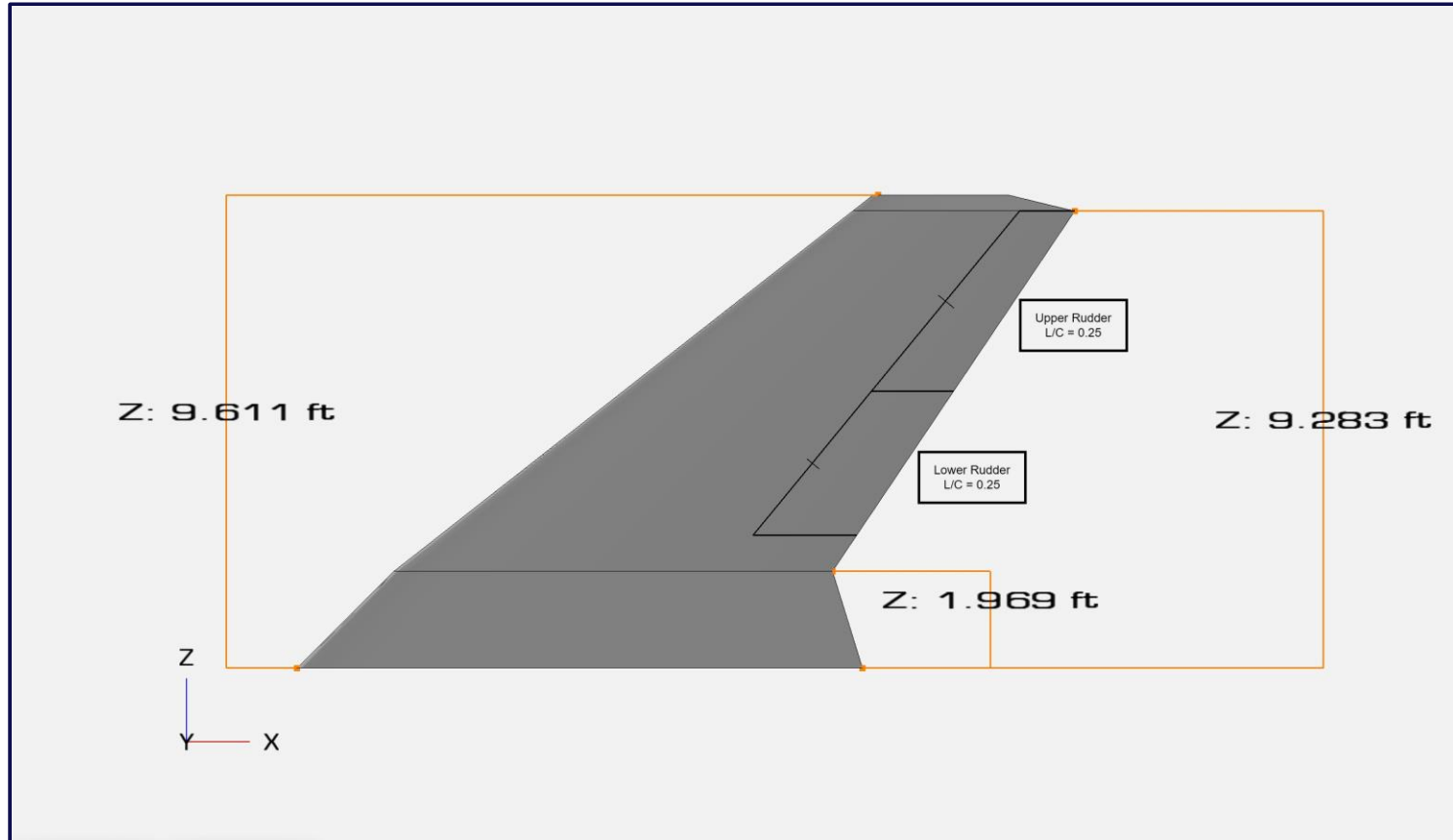
Control Surfaces



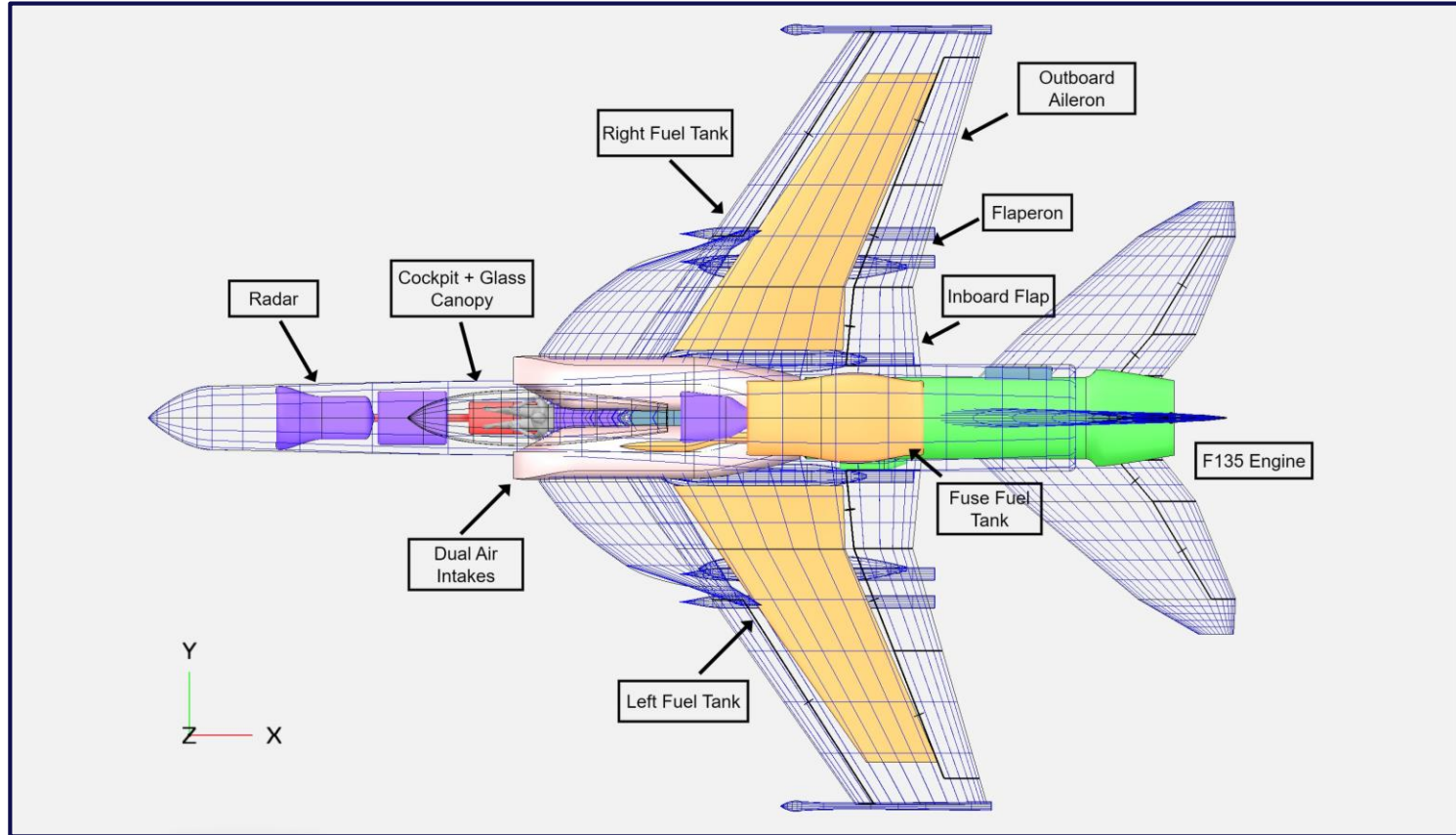
Empennage (Horizontal)



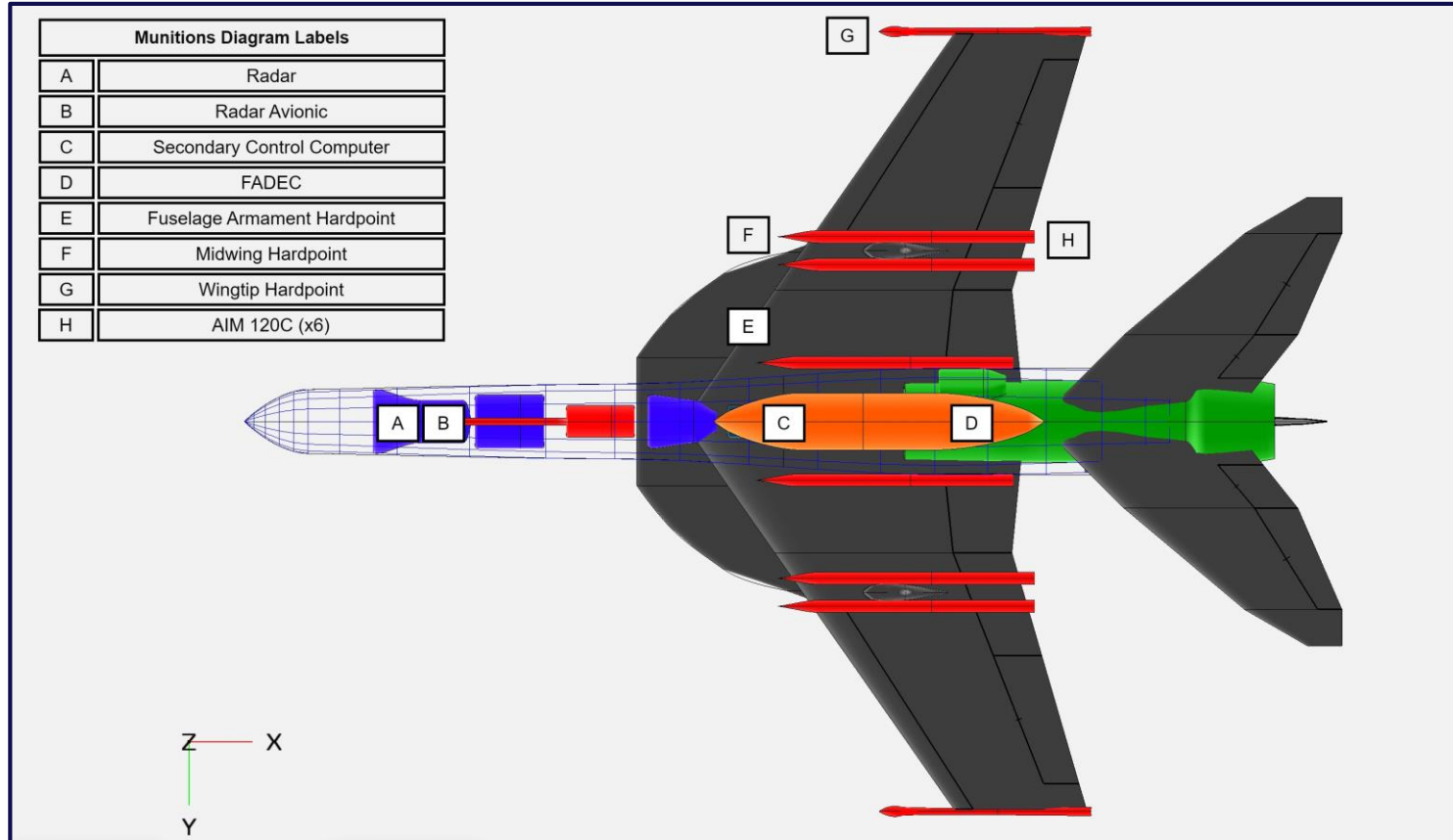
Empennage (Vertical)



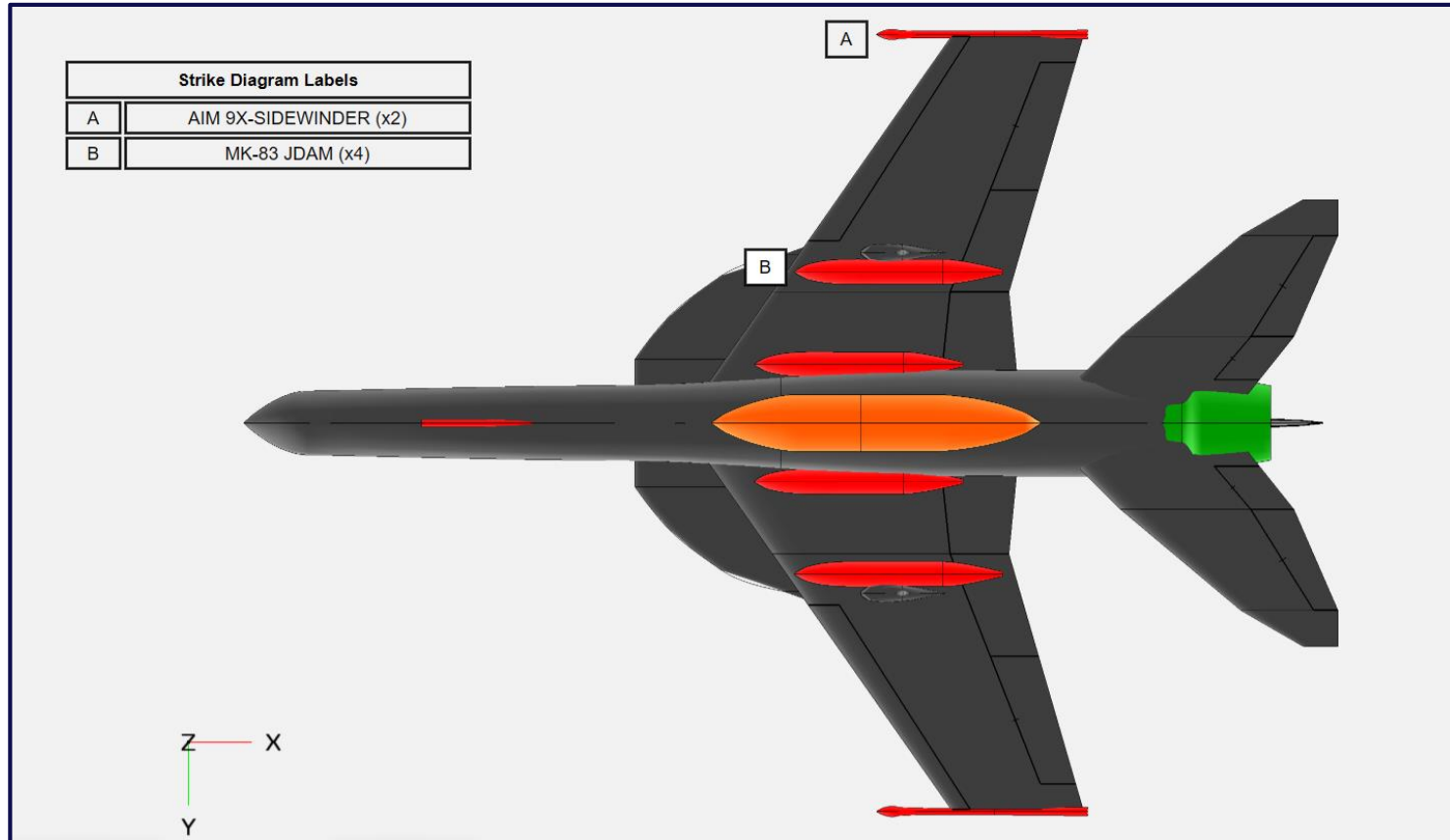
Internal Layout:



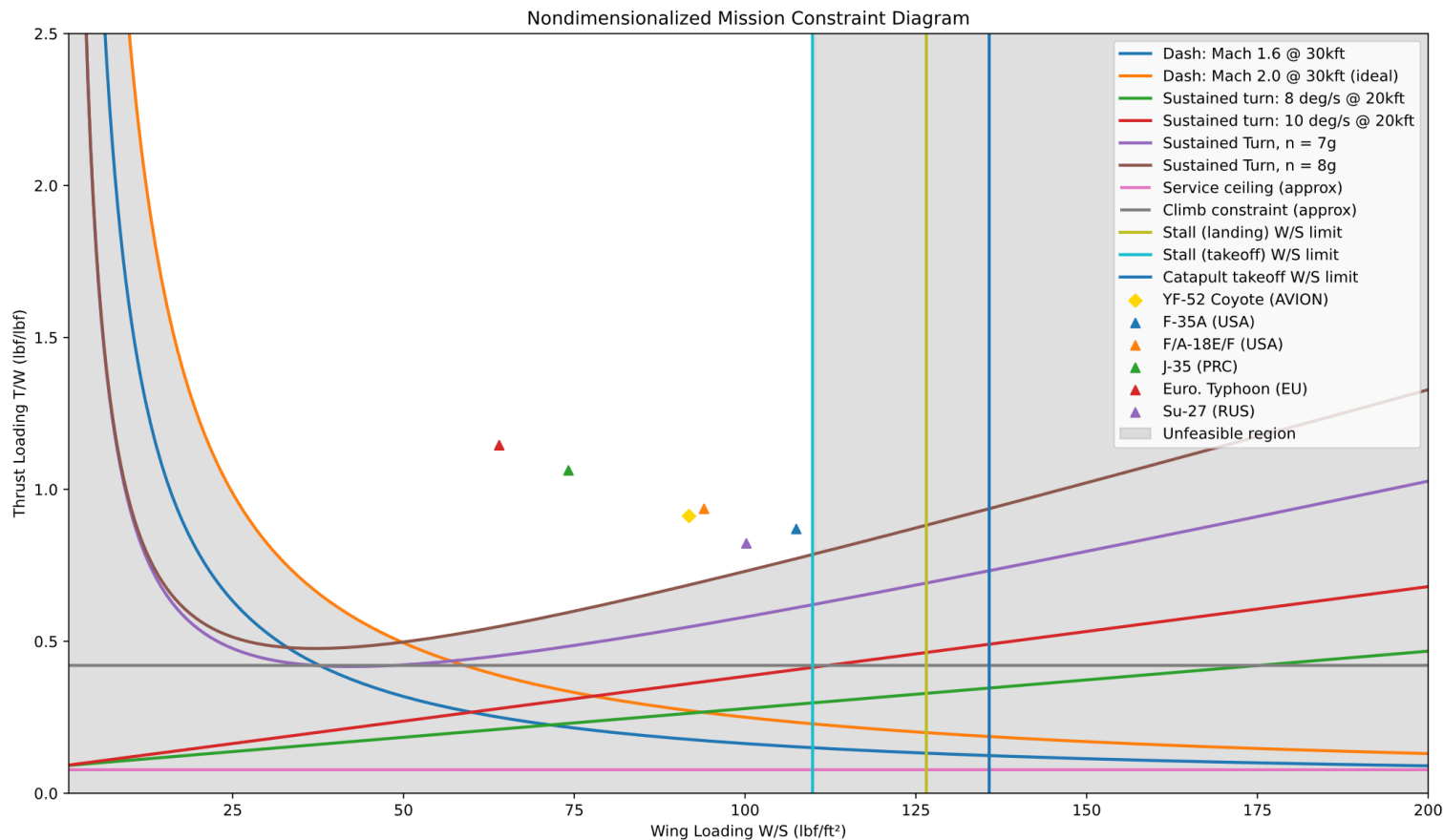
Air Superiority Mission Payload:



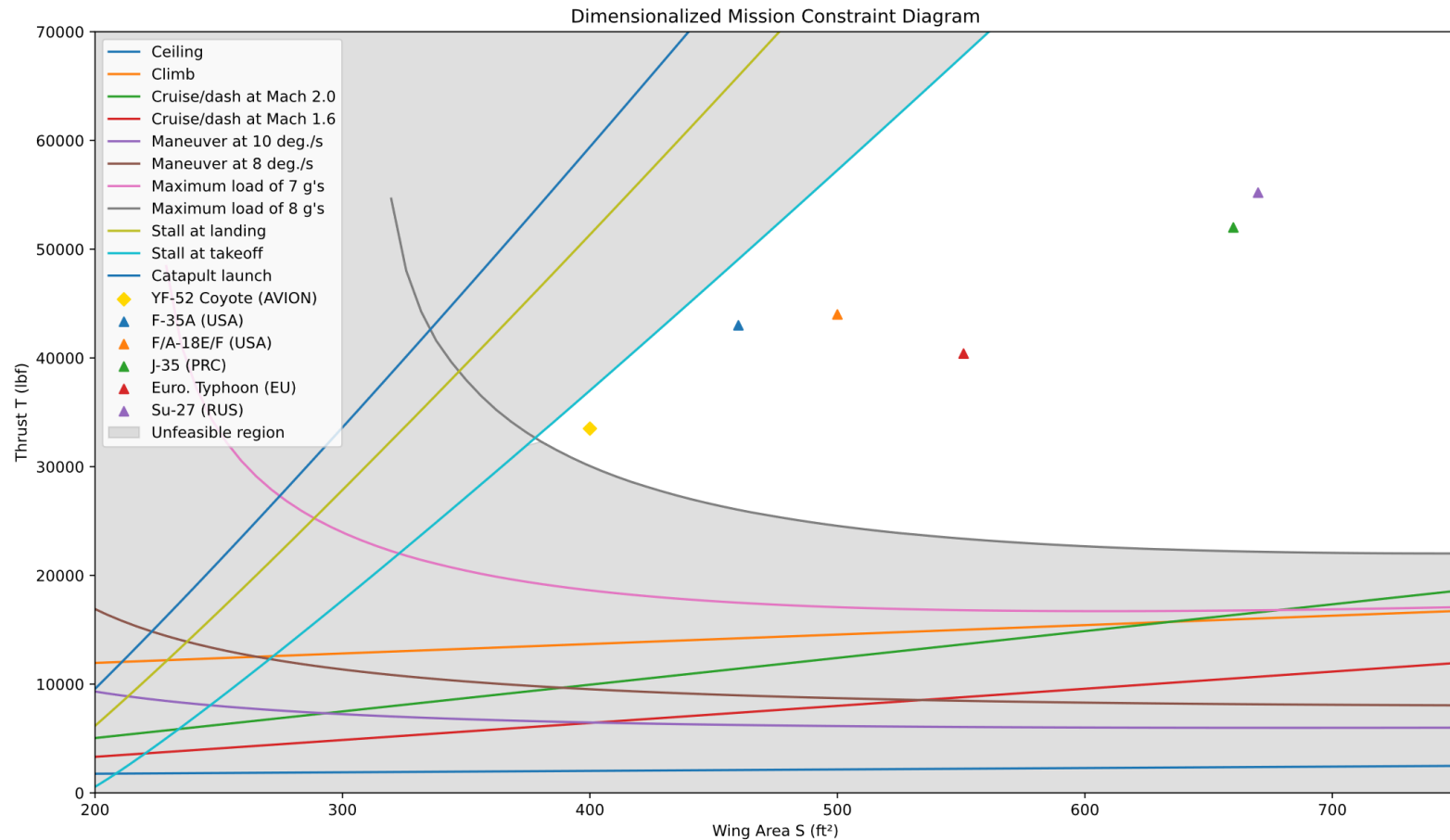
Strike Mission Payload:



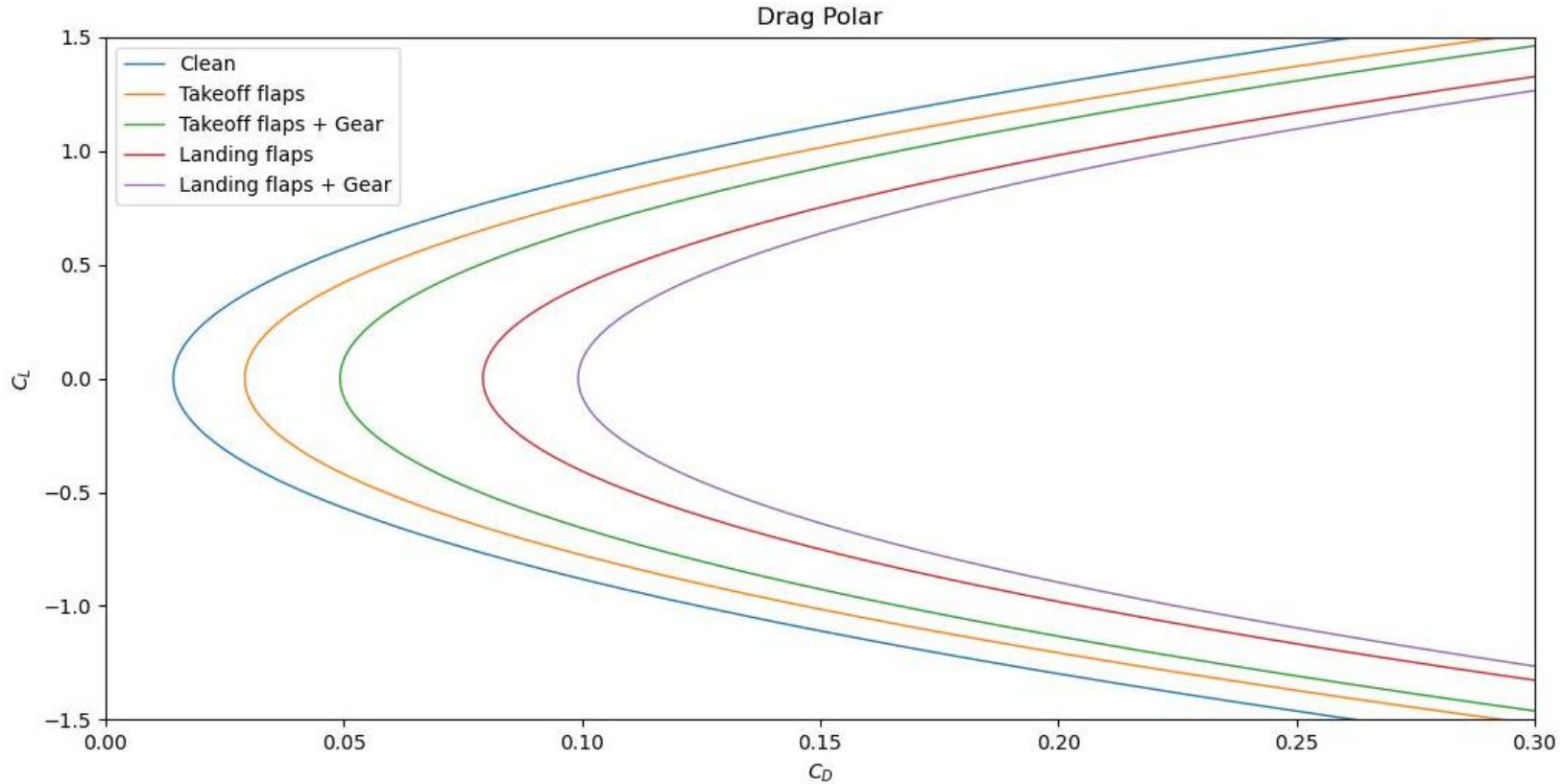
Preliminary Sizing:



Preliminary Sizing:



Drag Polar:



How our design meets RFP requirements

Single-engine configuration (F135 class)
reduces acquisition and life-cycle cost
compared to twin-engine aircraft

Higher thrust-to-weight ratio (~ 0.93)
improves maneuverability and acceleration

Combat radius ≥ 700 nmi achievable through
efficient aerodynamic configuration and fuel
fraction sizing

Sustained turn performance ≥ 8 deg/s meets
maneuverability requirements

Carrier compatible geometry that satisfies both
carriers (CVN-68 / CVN-78) operations

Mach 2 dash capability exceeds the required
Mach 1.6



Innovations

1. Strakes/ Leading-edge root extensions

Purpose:

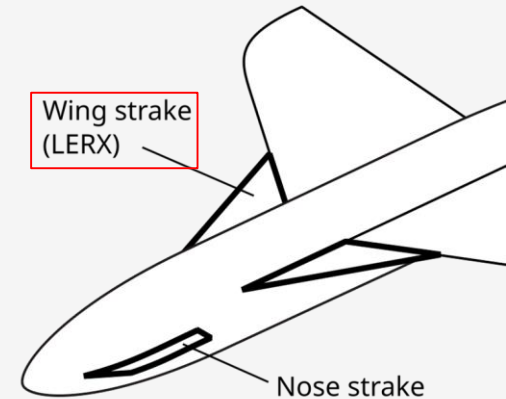
1. Generate Vortices at high AOA
2. Delay flow separation
3. Improve lift and maneuverability

Benefits:

1. Increased CL_{max}
2. Improved dogfight performance
3. Better carrier landing control

Aircrafts using similar system:

- F/A - 18
- Sukhoi Su-27
- F-16



Innovations

2. Diverterless supersonic Engine Intakes

Advantages:

1. Reduce weight
2. Lower drag
3. Improved stealth potential
4. Reduced maintenance costs
5. Fewer moving components

Aircrafts using similar system:

- F-35
- Sukhoi Su-75
- Chengdu J-20



Risks

Engine failure

- Higher consequences

Mitigation:

- Use of modern, reliable engine (F135)
- Redundant electrical and hydraulic systems

Engine Sizing

- Insufficient performance

Mitigation:

- Iterative constraint diagram analysis
- Benchmarking against existing fighter engines

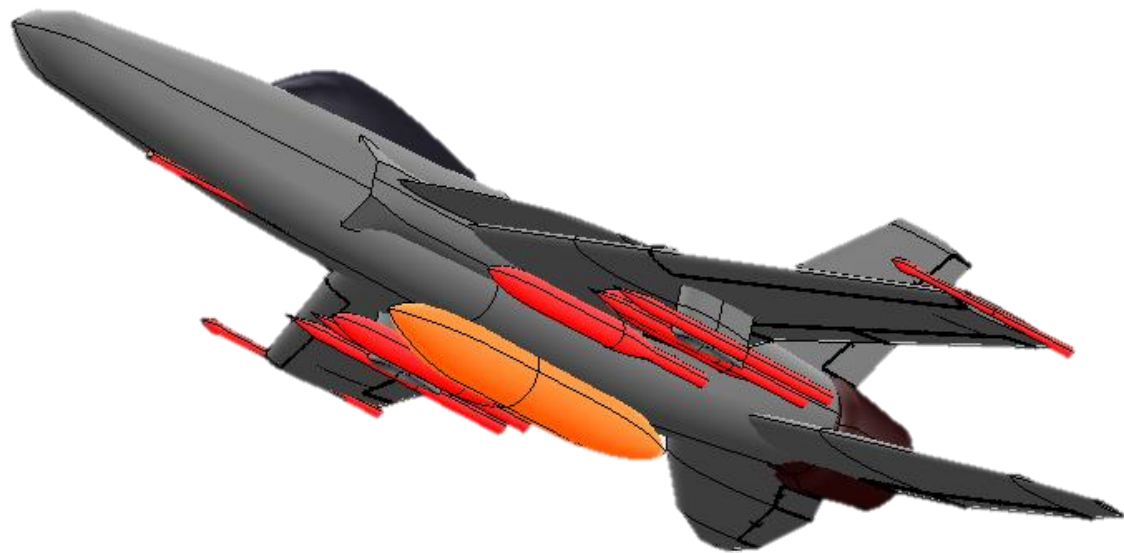
Auxiliary power unit

- Reduced survivability engine failure scenario

Mitigation:

- Prioritize APU sizing for enhanced survivability





Thank you.

